Signing the Windows installer (code-signing certificate)

Airline Manager Pro · planning notes · prices checked 2 November 2026

Why

Today the PC Setup (AirlineManagerPro-PC-Setup-x.y.z.exe) and the game exe are not signed. On the first run Windows shows **"Windows protected your PC"** (SmartScreen) with the publisher "Unknown publisher", and the player has to click *More info* → *Run anyway*. Signing puts the company's verified name on the files and proves they haven't been changed since they were built.

How it works

1. A certificate authority (or Microsoft) checks who the company is: business registration, address, phone, sometimes a D-U-N-S number.
2. It issues a signing certificate. Since June 2023 the private key must stay on secure hardware: a USB token, or a cloud signing service. It can no longer be delivered as a file.
3. Our Windows build signs both exes as its last step, after the icon, version and window settings are put in. With a cloud service this runs automatically in the GitHub build; the account details are kept in GitHub's secret settings, **never in the public repository**.
4. Windows then shows the company as the publisher.

The SmartScreen warning does not disappear at once

SmartScreen trusts a signature once enough people have downloaded and run files signed with it. With a new certificate the warning still shows for a while and fades as downloads add up. The trust stays with the certificate, so later versions signed with it start trusted. **EV certificates no longer skip this step**; they are only needed for Windows drivers, so they aren't worth the extra price here.

Options

Option	Price (about)	Who can get it	Notes
Microsoft Artifact Signing (formerly Trusted Signing)	about $10 a month (Basic, 5,000 signatures)	Companies in the USA, Canada, the EU and the UK; individuals only in the USA and Canada	Cheapest paid option; works straight from GitHub builds. A company in the Faroe Islands (outside the EU) does not qualify; a company registered in Denmark would.
SSL.com OV certificate + eSigner cloud signing	about $129 a year + eSigner from about $15 a month	Companies, sole traders, also individuals (IV)	Cheapest certificate authority found; cloud signing works in GitHub builds. About $300 a year in total.
GlobalSign OV	about $434 a year	Registered businesses	Token or cloud signing extra.

Option	Price (about)	Who can get it	Notes
DigiCert OV	about $700–840 a year	Registered businesses	Token or KeyLocker cloud signing.
Microsoft Store (MSIX package)	Free (individual developer accounts have no fee)	Anyone with a Microsoft developer account	Microsoft signs the package, so no SmartScreen warning. But the Store installs it: the custom installer (drive and folder choice, our uninstaller) isn't used for that version. Can be offered alongside the Setup exe from the website.
SignPath Foundation, Certum Open Source	Free / low	Open-source projects only	**Not usable**: the game is a commercial product.

A D-U-N-S number (free from Dun & Bradstreet) speeds up the checks at most certificate authorities.

Suggested plan

1. Register the company (the name is still being chosen).
2. If it is registered in the EU (for example Denmark): Microsoft Artifact Signing. Otherwise: SSL.com OV certificate with eSigner.
3. Put the signing account details in GitHub's secret settings; the build then signs the game exe and the Setup exe automatically on every release.
4. Optional: also publish an MSIX package in the Microsoft Store, for players who want no warning at all.

Sources

- Microsoft Q&A, Artifact Signing eligibility: https://learn.microsoft.com/en-us/answers/questions/5915697/can-an-irish-sole-trader-(non-incorporated)-qualif
- Azure Artifact Signing pricing: https://azure.microsoft.com/pricing/details/trusted-signing/
- OV/EV code-signing prices 2026, hardware key rule, SmartScreen: https://codenote.net/en/posts/ov-ev-code-signing-certificate-pricing-japan-2026/
- SSL.com eSigner pricing: https://www.ssl.com/guide/esigner-pricing-for-code-signing/

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